

Hatim Kanchwala

 7 February 1995
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Summary

Experienced RTL Engineer with a strong focus on system-level design. Contributed to Adreno GPU micro-architectures at Qualcomm, enhancing performance, timing and power consumption. Extensive work in Germany on real-time FPGA simulations and hardware implementations. Proficient in SystemVerilog and essential design tools. Seeking roles to address modern hardware design and verification challenges, aiming to develop solutions for next-generation computing.

Experience

Jan. 2023 – Present
Bengaluru, India 

RTL Design Engineer Qualcomm India Pvt Ltd

- Worked on micro-architecture design and SystemVerilog implementation of Adreno GPU blocks in mobile and compute tier SoC.
- Received appreciation for thorough and high-quality RTL designs and micro-architecture diagrams.
- Executed design cleanup tasks to improve code readability and maintainability across multiple projects.
- Proposed AI initiatives and built automation scripts in Python for auto-generation of SystemVerilog RTL, saving repetitive manual effort across multiple projects.
- Analyzed and implemented fixes for timing improvement, power optimization, and wire count reduction.
- Supported verification effort by providing whitebox coverage RTL and formal checks for functional behaviour.

Oct. 2020 – Dec. 2021
Aachen, Germany 

Research Engineer Institute of Energy and Climate Research (IEK-10) Forschungszentrum Juelich GmbH

- Developed design with soft-core microprocessors to rapidly prototype control-loop algorithms for FPGA-based real-time simulators.
- Implemented models using High-Level Synthesis designs for RTL co-simulation and real-time simulation.
- Conceptualised heterogeneous architecture of control and data-logger soft-cores dedicated to running control algorithms at switching frequency and logging simulation data.
- Extended HLS models with memory-mapped AXI4 register interface and verified on Virtex-7 VC707 FPGA board.
- Developed automation scripts on Linux for code generation, synthesis, and bitstream generation stages.

May 2019 – Sept. 2020
Aachen, Germany 

Research Engineer, Development of FPGA Systems Institute for Automation of Complex Power Systems E.ON Energy Research Centre

- Integrated Xilinx FPGA boards into VILLAS co-simulation platform by designing an architecture built on top of Aurora 8B/10B serial protocol.
- Engineered automation scripts for design generation and bitstream compilation leading to version-agnostic local toolchain use.
- Developed designs in Verilog/VHDL and driver programs in C/C++ for FPGA firmware.

May 2018 – Nov. 2018
Naval Physical &
Oceanographic Laboratory,
DRDO 

Senior Research Engineer "Underwater Target Motion Analysis with Passive Sensors"

- Implemented advanced tracking filters in MATLAB, like Distributed Extended Kalman Filter, Shifted Rayleigh Filter, and Sequential Monte Carlo methods, for the Bearings-only Tracking problem.
- Simulated performance of modern filters on real field manoeuvre data from Indian Navy, and prepared comparative study.
- Concluded Shifted Rayleigh Filter outperforms other filters in terms of computational complexity and tracking accuracy.

Aug. 2017 – May 2018
IIT Patna, India 🇮🇳

Research Engineer

“Hardware Architecture of a Family of Sigma-Point Kalman Filters for Bayesian Estimation”

Control and Instrumentation Lab

- Designed and implemented a parallel architecture of Sigma-point Kalman filtering algorithms on FPGA.
- Conceptualised novel parallel routine for Cholesky matrix decomposition; improvement from $O(N^3)$ to $O(N)$ time complexity. Customised and implemented a matrix inversion routine to compute covariance inverses in $O(N)$ time complexity.
- Optimised resource usage of Cholesky decomposition architecture for double utilisation at same processor count.
- Implemented parallel designs in Verilog HDL using Vivado and open-source IPs on Zynq-7000 ZC702 and Digilent Nexys4 DDR FPGA boards.

Projects

May 2017 – Aug. 2017
Google Summer of Code
2017 🌐

Software Developer

“EDSAC Museum on FPGA”

Free and Open Source Silicon Foundation

- Built Verilog model of historic EDSAC computer from original but incomplete documentation in collaboration with experts from The National Museum of Computing, UK.
- Programmed and simulated EDSAC architecture and ISA on myStorm Lattice iCE FPGA board using open-source toolchains, like Yosys and iverilog.
- Designed and implemented modified-UART communication protocol to support external embedded I/O interfaces for Hardware-in-Loop application.
- Demonstrated final work product at ChipHack 2017 workshop and presented at ORConf 2017 digital design conference in Hebden Bridge, UK. Received full sponsorship.

Apr. 2016 – Aug. 2016
Google Summer of Code
2016 🌐

Software Developer

Coreboot (Flashrom), “Read/Write Multiple Status Registers and Lock/Unlock Memory on SPI Chips”

- Designed unified model for abstraction of status registers, configuration bits, and memory protected areas in SPI Flash-memory chips across diverse chip manufacturers.
- Developed CLI to expose new features, and tested infrastructure using Raspberry Pi and microcontroller development board.
- Liaised with SPI chip manufacturers from China to arrange for engineering samples and add support for those chips.

Education

Apr. 2019 – Oct. 2022
Aachen, Germany 🇩🇪

M. Sc. Electrical Engineering, Information Technology and Computer Engineering

RWTH Aachen University

Final Grade 2,1 (German) = 8 / 10 (Indian)

Thesis: “Field-Programmable Gate Array based Real-Time Control and Simulation”

July 2014 – May 2018
Bihta (Patna), India 🇮🇳

B. Tech. Electrical Engineering

Indian Institute of Technology Patna

Final Grade 7.32 / 10

Thesis: “Hardware Architecture of a Family of Sigma-Point Kalman Filters for Bayesian Estimation”

- Nominated for Best B. Tech. Thesis award from Dept of Electrical Engineering.
- One of only two students to receive 10 / 10 grade.

2001 – 2011
Aurangabad, India 🇮🇳

Class X Central Board of Secondary Education

Nath Valley School

Final Grade 10 / 10

Skills

Programming	SystemVerilog/VHDL, C/C++, Python, Java, Tcl, JavaScript, Shell				
Software	Verdi, Xilinx Vivado & HLS, MATLAB, Simulink, git/GitHub, RSCAD, NI LabVIEW, GNU/Linux, gdb, Verilator, yosys, L ^A T _E X, gnuplot				
Hardware	Xilinx Virtex & Zynq SoC, Digilent Nexys4 DDR, Lattice FPGA, Embedded Microcontrollers				
Languages	English	<i>Listening C2</i>	<i>Reading C2</i>	<i>Speaking C2</i>	<i>Writing C2</i> <i>TOEFL 114 / 120</i>
	Deutsch	<i>Hören B2</i>	<i>Lesen B2</i>	<i> Sprechen B1</i>	<i>Schreiben B1</i>
	Hindi	native			